ALEX SALAMATI

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EDUCATION

Loyola Marymount University *Los Angeles, CA | Expected May 2027*
B.S. Computer Engineering, Minor: Computer Science
Relevant Coursework: Data Structures & Algorithms, Microprocessors/Microcontrollers, Computer Architecture (SystemVerilog), Circuits I & II, Electronics, Digital Systems | Fall 2026: Embedded Systems, Machine Learning, Senior Capstone

TECHNICAL SKILLS

Languages: Python, C, C++, MATLAB, Bash
Robotics & Vision: OpenCV, MediaPipe, LeRobot, Imitation Learning (ACT), PID Control, Servo Control (PWM/serial bus), Teleoperation
Embedded & Hardware: Arduino, Raspberry Pi, dsPIC33, RTOS (Salvo), I2C, SPI, UART, Oscilloscopes, Logic Analyzers, 3D Printing/CAD
Tools: Git/GitHub, Linux, GNU Radio, SDR (USRP), Vivado, MPLAB X

SELECTED PROJECTS

SO-ARM101 Robot Arm — Imitation Learning *LeRobot | ACT | Python | 3D Printing* *Apr 2026 – Present*

- Assembled and calibrated a 6-DOF SO-ARM101 robot arm (3D-printed components, serial servo bus control); achieved working leader-follower teleoperation through the LeRobot framework
- Building imitation-learning pipeline: collecting teleoperated pick-and-place demonstrations to train an ACT policy for autonomous manipulation

Ball Balancing Robot *PID Control | Computer Vision | Arduino* *Mar 2026 – Present*

- Designing a closed-loop 3-DOF servo platform: OpenCV vision pipeline (30+ FPS) estimates ball position/velocity, feeding a PID controller driving servo PWM via Arduino
- Tuning PID gains (Kp, Ki, Kd) with step-response analysis to minimize overshoot and achieve stable, low-latency tracking

Real-Time Hand Gesture Tracker *MediaPipe | OpenCV | Python* *May 2025 – Jul 2025*

- Built real-time hand landmark detection classifying raised fingers from landmark coordinate geometry at 30+ FPS
- Rendered live finger-count overlay on video feed; architecture designed as a modular input layer for gesture-driven robot control interfaces

MIPS Single-Cycle Processor *SystemVerilog | FPGA | Vivado* *Mar 2026 – May 2026*

- Designed a complete 32-bit single-cycle MIPS processor in SystemVerilog; synthesized on Xilinx FPGA
- Wrote directed testbenches and verified correct execution across R/I/J-type instructions; analyzed timing to confirm setup/hold constraints

RESEARCH EXPERIENCE

CubeSat Communications Subsystem — Undergraduate Researcher *LMU | Advisor: Dr. Gustavo Vejarano* *May 2026 – Jun 2026*

- Developed satellite-side firmware (C, Salvo RTOS on dsPIC33) for an RF jamming-detection CubeSat; achieved verified end-to-end downlink — LOG, TEL, and IMG telemetry transmitted via AX.25 at 435 MHz to a GNU Radio/USRP ground station
- Designed binary-semaphore bus arbitration across concurrent RTOS tasks; debugged three independent faults (race condition, protocol error, GPS UART blocking) through single-variable testing
- Conducted byte-survival study confirming 97-byte payload ceiling and hex-encoding requirement for all image data

EXPERIENCE

IT Support Technician *LMU ITS Classroom Support | Work Study* *Jun 2024 – Present*

- Maintained and troubleshot AV, OS, and hardware systems across 20+ smart classrooms; diagnosed and resolved failures in under 5 minutes during live lectures
- Debugged cross-platform issues across Windows and macOS; documented recurring hardware failure patterns and escalated firmware-level issues to vendors

LEADERSHIP & ACTIVITIES

Member *Robotics Optimization & Building Organization (ROBO), LMU* *2023 – Present*

- Collaborated on robotics builds integrating sensors, motors, and embedded control; contributed to autonomous obstacle-avoidance project using ultrasonic sensors

Vice President *Additive Manufacturing Club, LMU | Member: IEEE Student Chapter* *2023 – Present*

- Led CAD modeling and 3D print optimization — directly applied to fabricating robot arm components